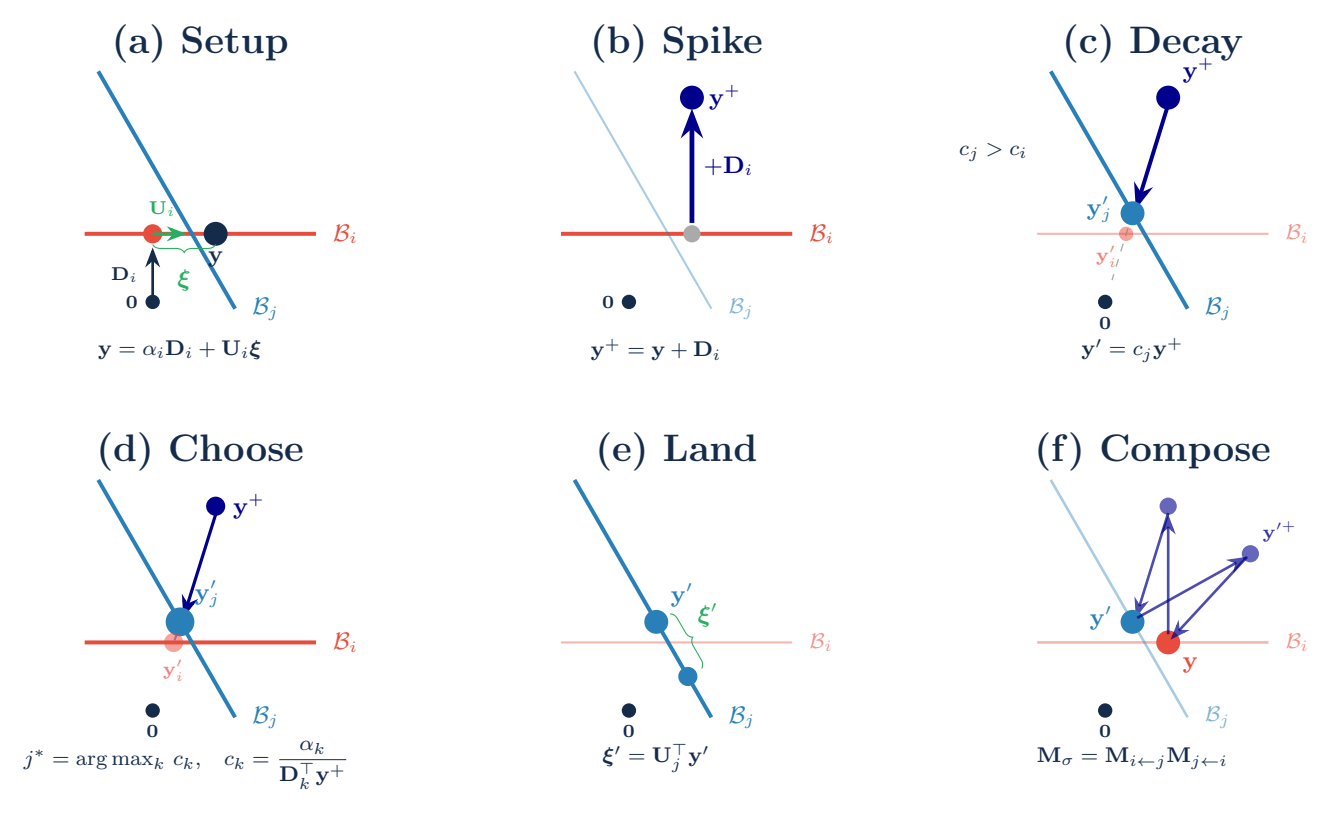

(a) Setup
$\mathbf{U}_i$
$\mathcal{B}_i$
$\mathbf{y}$
$\mathbf{D}_i$
$\boldsymbol{\xi}$
$\mathbf{0}$
$\mathcal{B}_j$
$\mathbf{y} = \alpha_i \mathbf{D}_i + \mathbf{U}_i \boldsymbol{\xi}$
(b) Spike
$\mathbf{y}^+$
$+\mathbf{D}_i$
$\mathcal{B}_i$
$\mathbf{0}$
$\mathcal{B}_j$
$\mathbf{y}^+ = \mathbf{y} + \mathbf{D}_i$
(c) Decay
$\mathbf{y}^+$
$c_j > c_i$
$\mathbf{y}'_j$
$\mathcal{B}_i$
$\mathbf{y}'_{il}$
$\mathbf{0}$
$\mathcal{B}_j$
$\mathbf{y}' = c_j \mathbf{y}^+$
(d) Choose
$\mathbf{y}^+$
$\mathbf{y}'_j$
$\mathcal{B}_i$
$\mathbf{y}'_i$
$\mathbf{0}$
$\mathcal{B}_j$
$j^* = \arg\max_k \, c_k, \quad c_k = \dfrac{\alpha_k}{\mathbf{D}_k^\top \mathbf{y}^+}$
(e) Land
$\mathbf{y}'$
$\boldsymbol{\xi}'$
$\mathcal{B}_i$
$\mathbf{0}$
$\mathcal{B}_j$
$\boldsymbol{\xi}' = \mathbf{U}_j^\top \mathbf{y}'$
(f) Compose
$\mathbf{y}'^+$
$\mathbf{y}'$
$\mathcal{B}_i$
$\mathbf{y}$
$\mathbf{0}$
$\mathcal{B}_j$
$\mathbf{M}_\sigma = \mathbf{M}_{i \leftarrow j} \mathbf{M}_{j \leftarrow i}$